Contamination models: declared failure modes for belief fusion

Robust belief fusion only earns its keep when some agents are wrong.

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Contamination models (part 2)

The active-inference community has built ensembles that coordinate by sharing beliefs and observations (Friston et al. 2024; Heins et al. 2024; Albarracin et al. 2022; Kaufmann et al. 2021),

Contamination models (part 3)

but it has assumed those beliefs are trustworthy in the cited modeled protocols: fusion is treated as exact-Bayes pooling of well-calibrated reports.

Contamination models (part 4)

The robust-Bayes and federated-learning literatures (McMahan et al. 2017; Ashman et al. 2022; Mildner et al. 2025) have, in turn, studied robustness to corrupted clients under their declared settings, but outside the generative-model-bearing POMDP setting.

Contamination models (part 5)

This section defines the corruption process that lets us test fusion robustness inside the active-inference colony — the experimental complement of the robust aggregation rule of sec. 8.8.

Corruption process for adversarial belief broadcasts

In the sentinel world a healthy sentinel reports a well-calibrated categorical over the creature's location; a contaminated one reports something corrupted. `contamination.contaminate` manufactures the corrupted reports.

Corruption process for... (part 2)

Every corruption is a convex mixture of the agent's belief b with a corruption target t, governed by a single rate $r \in [0, 1]$,

Corruption process for... (part 3)

$$\tilde{b} \;=\; (1-r)\,b \;+\; r\,t, \tag{21}$$

Corruption process for... (part 4)

so the experiments sweep exactly one knob. The convex form of eq. 21 gives a clean limit and is the anchor of the suite (ISC-26): at $r = 0$ every corruption kind returns the input belief unchanged, so contamination is a strict, continuous departure from the uncorrupted Friston belief-share —

Corruption process for... (part 5)

never a discontinuity.

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Corruption process for... (part 6)

This section defines the three core corruption targets t, each capturing a distinct failure of a federated agent. Geometrically the three are three landmarks of the probability simplex — a wrong vertex (`confident_wrong`), the flat centroid (`uniform`), and a random interior point (`label_noise`) —

Corruption process for... (part 7)

so the mixture of eq. 21 drags an honest belief toward a qualitatively different destination in each case.

Corruption process for… (part 8)

Two further mechanisms (`byzantine` and `drift`) extend the same convex-mix contract and are introduced in the extended-methods supplement (sec. 16.2).

Corruption process for... (part 9)

`confident_wrong` — **the adversarial sentinel.** This is the lookout that points to one wrong cell and insists on it with total certainty. The target is a one-hot spike on a wrong state, $t = \mathrm{onehot}(s_{\text{wrong}})$, so $\tilde{b}$ is mixed toward a confident, mistaken delta.

Corruption process for... (part 10)

Callers choose s_{wrong} explicitly; the verdict sweep of sec. 8.22 fixes it once per colony as the state diametrically opposite the true state on the location grid, held constant across the entire rate sweep, rather than deriving it from the agent's current belief.

Corruption process for... (part 11)

At $r = 1$ this is a pure delta on the wrong cell. This is the saboteur that is *sure* and *mistaken*: exactly the agent that robust aggregation must reject.

Corruption process for… (part 12)

`label_noise` — **the miscalibrated sentinel.** This is the lookout with a scrambled sensor: it is not lying toward any particular cell, only diluting every honest report with the same fixed sprinkle of noise. The target is a fixed noisy categorical drawn once from a $\mathrm{Dirichlet}(1)$ (a random but valid pmf),

Corruption process for... (part 13)

modeling a sentinel whose report is partly random rather than adversarial.

Corruption process for... (part 14)

Because the noisy target is drawn once and then held fixed across the rate sweep, the corruption has no direction to exploit and no single cell to veto — the robust pool meets diffuse degradation, not a targeted attack.

Corruption process for… (part 15)

`uniform` — **the apathetic sentinel.** This is the lookout that shrugs: it has lost track of the creature and calls every cell equally likely. The target is the maximum-entropy uniform pmf $t = (1/n_s)\mathbf{1}$, modeling a saturated sentinel that has lost all information. At $r = 1$ it reports uniform,

Corruption process for... (part 16)

contributing no evidence to the pool rather than actively pulling it toward a wrong cell.

Corruption process for… (part 17)

All three share the contract of eq. 21 and require an explicit seeded generator — `label_noise` uses it to draw the noisy target — so every contaminated report is reproducible. The grid of rates the sweep uses, $\{0, 0.225, 0.45, 0.675, 0.9\}$, deliberately stops below the pure-veto limit $r = 1$,

Corruption process for... (part 18)

where a fully-confident wrong delta forces *every* pooling rule's accuracy to zero and the robust-versus-naive contrast vanishes.

How contamination meets the three robustness axes

The authoritative guarantee taxonomy is sec. 6.3; this section only records where the declared corruption enters each route.

Client bounded-loss route

A corrupted local observation enters the per-agent generalized-Bayes update of sec. 8.7. FedGVI's bounded-influence result applies only under the source theorem's matching loss, divergence, model, and regularity assumptions (Mildner et al. 2025).

Client bounded-loss route (part 2)

The logistic-regression comparison in fig. 21 is a narrower exploratory proxy: it compares RCCE and NLL point-estimate gradients under two L2 coefficients. Its legacy `AR` argument selects the larger coefficient;

Client bounded-loss route (part 3)

it does not evaluate Alpha-Rényi divergence or inherit the source theorem from the observed curve.

Heuristic server route

A corrupted posterior broadcast enters `robust_aggregate` at fusion. Divergence reweighting can assign a smaller effective weight to a report far from the current consensus, as the finite diagnostics figs. 16, 18 illustrate.

Heuristic server route (part 2)

The rule's positive formal property is the exact $c = 0$ recovery limit eq. 10, not a client-loss bound, Byzantine guarantee, or universal robustness theorem.

Objective-backed variational server route

The same corrupted posterior can instead enter `variational_aggregate`, whose declared free energy, block-descent property, and effective-weight bound are given in sec. 8.8.2.

Objective-backed variational... (part 2)

Those properties belong to that forward-oriented objective and do not certify `robust_aggregate` or the source client's bounded-loss theorem.

Objective-backed variational... (part 3)

Contamination is therefore a common stressor, not a license to transfer a guarantee or empirical result between client, heuristic-server, and variational-server lanes.